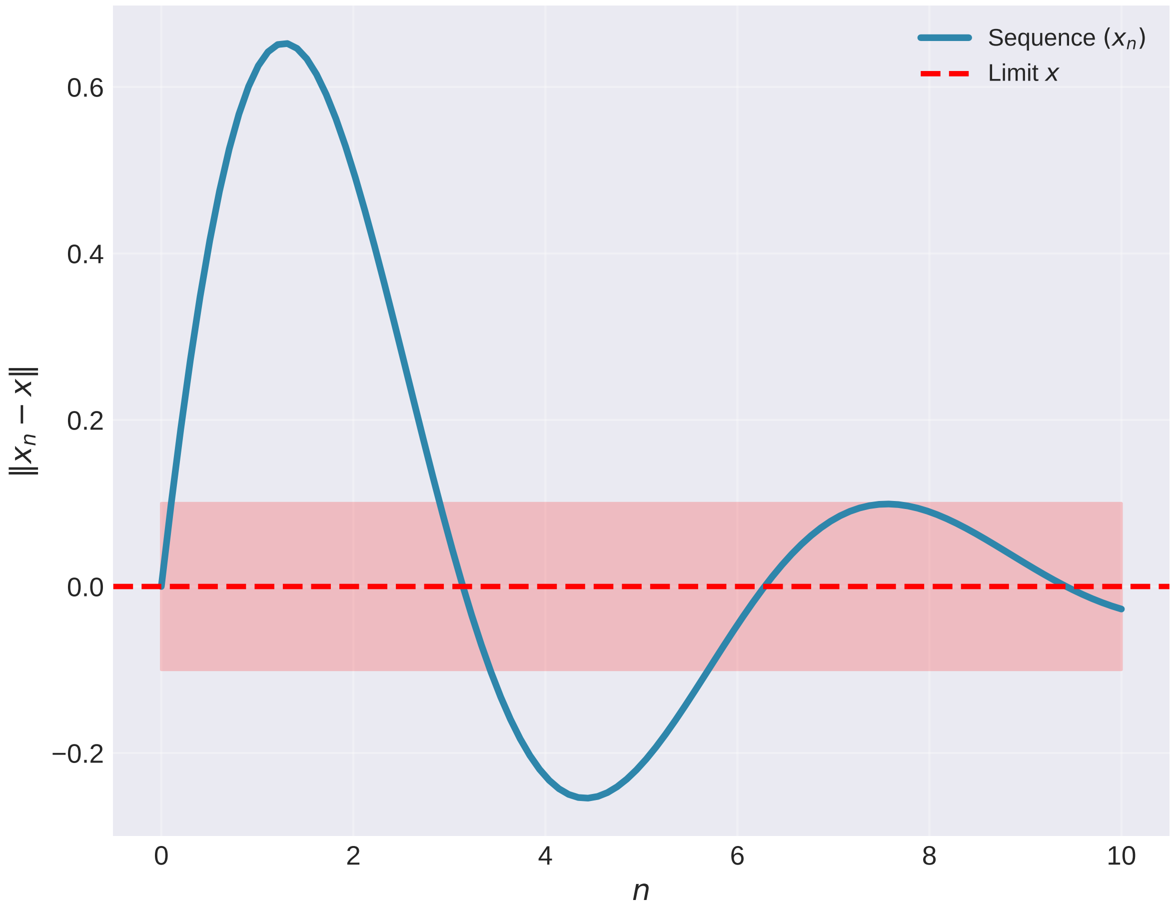
Key Concepts in Banach Spaces

Strong Convergence in Norm



Linear Operator

$$T : X \to Y$$

$$\boxed{T(\alpha x + \beta y) = \alpha T(x) + \beta T(y)}$$

Bounded if: $\|T\| = \sup_{\|x\|=1} \|T(x)\| < \infty$

Dual Space X^*

Space of bounded linear functionals

$$f : X \to \mathbb{R} \text{ or } \mathbb{C}$$

Riesz Representation Theorem:

For $f \in \mathcal{H}^*$, $\exists!\, y \in \mathcal{H}$:

$$\boxed{f(x) = \langle x, y \rangle \; \forall x \in \mathcal{H}}$$

Weak Convergence

$x_n \rightharpoonup x$ *(weak)*

$\lim_{n\to\infty} f(x_n) = f(x)$ for all $f \in X^*$

vs.

$x_n \to x$ *(strong)*

$$\lim_{n\to\infty} \|x_n - x\| = 0$$